A. MEXanized Array

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: constructive algorithms, greedy, math | https://codeforces.com/problemset/problem/1870/A

You are given three non-negative integers n, k, and x. Find the maximum possible sum of elements in an array consisting of non-negative integers, which has n elements, its MEX is equal to k, and all its elements do not exceed x. If such an array does not exist, output -1.

The MEX (minimum excluded) of an array is the smallest non-negative integer that does not belong to the array. For instance:

- The MEX of [2,2,1] is 0, because 0 does not belong to the array.
- The MEX of [3,1,0,1] is 2, because 0 and 1 belong to the array, but 2 does not.
- The MEX of [0,3,1,2] is 4, because 0, 1, 2 and 3 belong to the array, but 4 does not.

Input

The first line contains a single integer t ($1 \le t \le 1000$) — the number of test cases. Then follows the description of the test cases.

The only line of each test case contains three integers n, k, and x ($1 \le n, k, x \le 200$).

Output

For each test case, output a single number — the maximum sum of elements in a valid array, or -1, if such an array does not exist.

Example

Input

```
9
5 3 3
4 7 5
4 2 28
12 10 6
57 51 122
200 1 200
2 2 1
3 2 1
4 7 10
```

Output

```
7
-1
57
-1
2007
39800
1
2
-1
```

Note

In the first test case, the maximum sum is 7, and one of the valid arrays is [0, 1, 2, 2, 2].

In the second test case, there are no valid arrays of length n.

In the third test case, the maximum sum is 57, and one of the valid arrays is [0, 1, 28, 28].